

Wildlife biology and computer science student specializing in conservation technology, with experience integrating UAS, GIS, and machine learning for ecosystem monitoring.

Education

University of Montana, Missoula, Montana
Bachelor of Science in **Wildlife Biology (Terrestrial)**
Bachelor of Science in **Computer Science (Data Science)**
Certificates: Global Leadership Initiative, Geographic Information Systems, Honors

Expected Graduation: May 2027
Phi Kappa Phi Member (inducted 2023)
Current GPA: 3.86

Research and Field Experience

Lead Intern Autonomous Aerial Systems Office, University of Montana

May - August 2025

- Managed team of 9 interns across 3 concurrent projects (GPS survey, campus modeling, rangeland assessment), coordinating workflows from field data collection through quality control to deliver geospatial products.
- Established 200+ 3D ground control points using RTK GPS technology and produced georeferenced 2D maps and 3D models using Pix4D and ArcGIS Pro for feature extraction and planning future data collection efforts.
- Conducted multispectral drone surveys of 550 acres of rangeland to explore vegetation assessment methodologies and gathered 2 river miles of thermal imagery for river connectivity monitoring.

Wildlife Internship

May - August 2024

Field Study Abroad Program, Limpopo, South Africa

- Gained practical field experience in wildlife management and monitoring, including field ranger training for approaching dangerous wildlife, driving all-terrain vehicles, and identifying wildlife tracks with professionals.
- Developed skills in conservation strategies through exposure to international best practices and community-based approaches in wildlife management.
- Expanded understanding of endangered species conservation by examining the interconnected challenges of poaching, poverty, and habitat loss in developing countries, gaining insights into how socio-economic factors influence wildlife management and the importance of community-centered conservation efforts to address these issues sustainably.

Research Lab Assistant - 4-5 hours per week

January 2023 - June 2024

Charismatic Minifauna Lab, Forestry Bldg, Missoula, MT 59812
erim.gomez@mso.umt.edu

- Collaborated with researchers to develop a training dataset of over 1000 labeled spectrogram samples for CNN model training and validation.
- Identifying bird calls from 100s of spectrograms generated from acoustic data using Raven Pro software.
- Applied machine learning to wildlife monitoring by training a CNN classifier to automate identification of cuckoo vocalizations from acoustic datasets, reducing manual analysis time and costs.

Bird Ecology Field Technician - approximately 48 hours per week

May 2023 - July 2023

- Conducted point-count, playback, and vegetation surveys for black-billed and yellow-billed cuckoos in a remote riparian ecosystem. Utilized GPS to navigate trailless terrain outside of cell service.
- Programmed and installed autonomous audio recorders (SongMeter, AudioMoth).

Technical Skills and Trainings

- Certifications:** Level II Track and Sign CyberTracker Certification (87%), Level III in South Africa, **Federal Aviation Association Part 107 Remote Pilot License**
- Software & Analysis:** Python, R, Machine Learning (CNN), ArcGIS Pro/Online, Pix4D, Google Earth Engine, Raven Pro
- Field Equipment:** Manual Transmission Vehicles, Wingtra, DJI platforms, Skydio, Emlid Reach RS2+, AudioMoth, SongMeter
- Trainings:** Wolf trapping and radio telemetry, Wilderness First Aid Training

Additional Work Experience

Database Project Manager | State of Montana Arboretum

November 2023 - Present

Missoula, MT 59812

- Collaborating with a diverse management team and 3 partner organizations to identify and ameliorate an ArcGIS Pro Online, web-based database of 2000+ records.
- Effectively communicating technical database and data storage processes to non-technical audiences.
- Consolidate imagery and feature data, enter geospatial data, and visualize data in web-geodatabase
- Establish protocol for database management as project manager

Relevant Coursework

Research Design	Machine Learning	Data Science Analytics	General Ecology
Conservation of Wildlife Populations	Unmanned Aerial Systems Regulations & Surveying	Applications of GIS	Data Visualization

Leadership and Service

- Global Leadership Initiative Board Member - Alumni Representative
- August 2025-Present
- Advocate for student perspectives on board, informing curriculum design decisions and program improvements
 - Contribute to leadership curriculum development through feedback on course content, experiential learning, and cross-cultural competency programming
- Global Leadership Initiative Student Ambassador | November 2024-May 2025
- Promoted GLI events, represented program to donors and stakeholders, and provided peer advising

Professional Memberships: Phi Kappa Phi | Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) | The Wildlife Society, Missoula Chapter | Honors Student Association

Leadership Roles: Founding President, Origami Club | Former Founding President, Computer Science Club | Former Secretary, International Student Association

Recognitions and Awards

- Benjamin A. Gilman International Scholarship, 2023
- Experiential Learning Scholarship, 2023, 2024, University of Montana
- Irene Evers Scholarship for Undergraduate Research, 2023, University of Montana
- UM Housing Student Staff Excellence Award, 2023, University of Montana
- Student Housing Staff Excellence Award, 2023, Diversity Advisory Council, University of Montana
- SACNAS NDiSTEM Travel Scholarship Award, 2022
- Student Diversity Leadership Award, 2022, Diversity Advisory Council, University of Montana
- Academic Achievement Scholarship 2021-present, University of Montana
- Dean's List, Fall 2021 - Spring 2024, University of Montana

Professional Development

- Montana Wildlife Society Conference, 2023-24
- Centering Indigenous Knowledge Webinar, 2022-24
- Stolen Waters Summit, 2023, University of Montana
- National Wilderness Workshop (Society for Wilderness Stewardship), 2023
- North American Wildlife Tracker Conference, 2023
- National SACNAS NDiSTEM Conference, 2022

Additional Information

- Languages: Native English speaker, Fluent in conversational French
- Field Skills: Wildlife tracking, hiking, cross-country skiing, snowshoeing, animal origami
- International Experience: South Africa, Sri Lanka (3x), France, Québec